



AI Infrastructure Market Disruptors in 2026

Emerging Market Dynamics Team, HTI Research

Nine Primary AI Infrastructure Disruptors Starting in 2026

Market Economics

- AI infrastructure demand will grow through 2030 and will remain in excess of supply.
- ODM selection will make or break product roadmap advantages for AI infrastructure providers.
- **By 2030, new devices will market AI as a hardware, functional foundation, not a feature.**

Technology

- **Co-packaged optics will be mainstream in AI data centers in three years.**
- Data centers will adopt on-site power or fail.
- 800VDC AI data centers launch in 2027: SiC and GaN become the power backbone.

Geopolitics

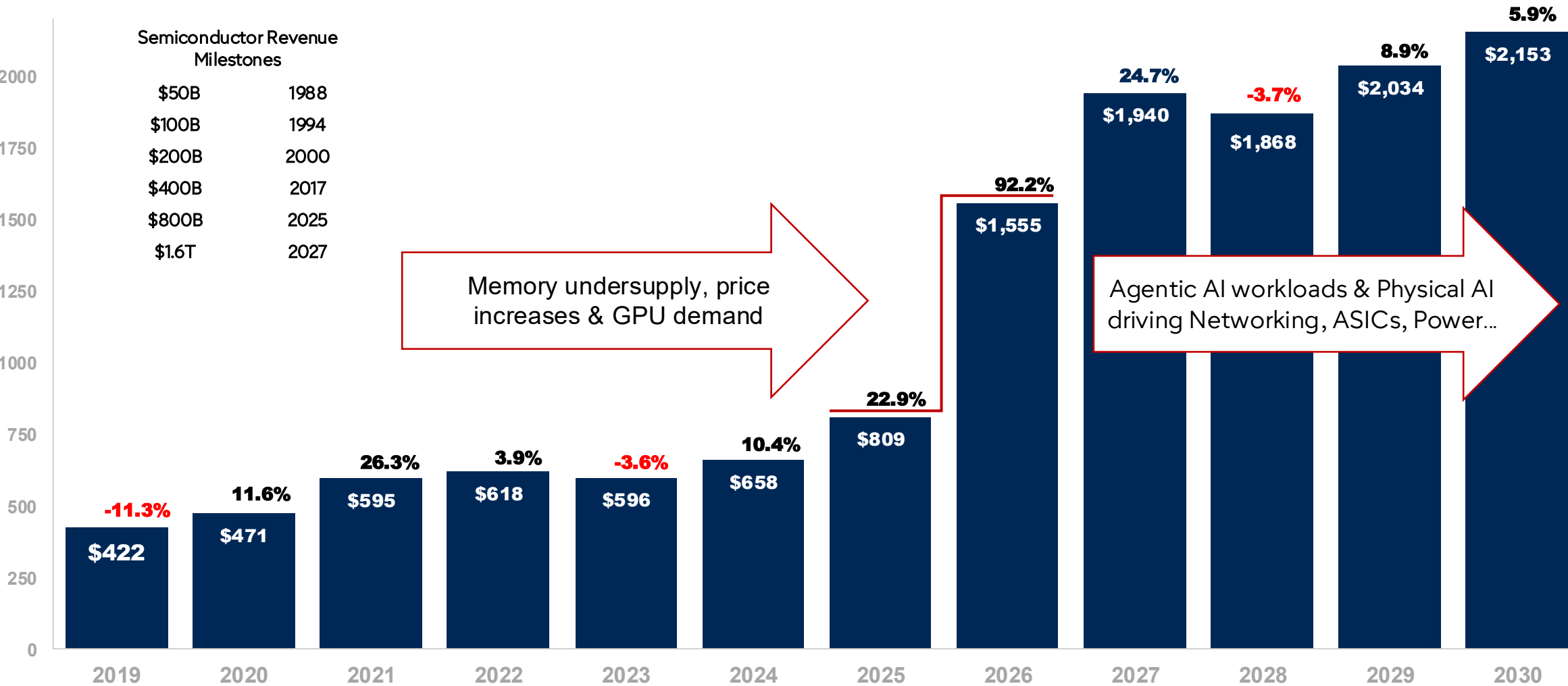
- China will exceed 50% semiconductor self-sufficiency by 2030.
- **By 2028, more than 75% of all enterprises outside of the U.S. will have a digital sovereignty strategy.**
- Taiwan is the AI infrastructure chokepoint.

1. How will AI impact electronic equipment features at the end of the decade?
2. Why will co-packaged optics be mainstream in datacenters in three years?
3. How broad is the Sovereign AI opportunity?

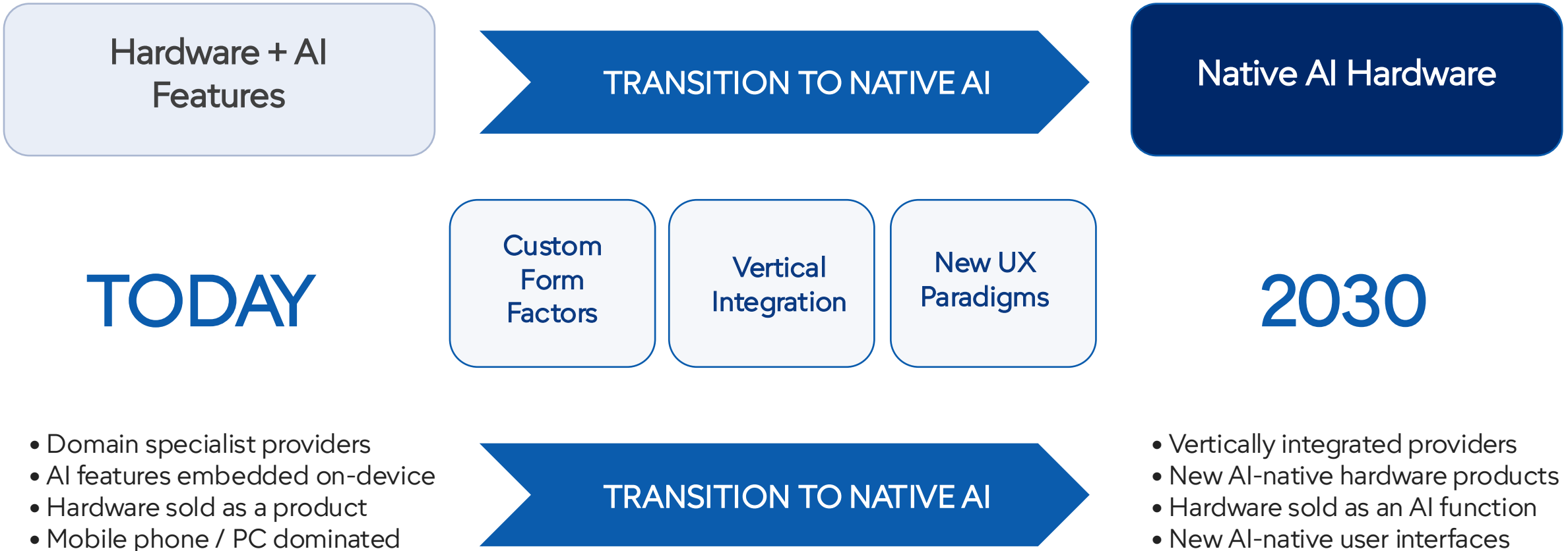
2026 Marks a Step Change in Semiconductor Revenue Growth

Billions of Dollars and Revenue Growth

■ 2Q26 Forecast



AI Will Define Hardware, Not Merely Enhance It



Source: [Market Trend: Edge AI Semiconductor Decisions Define Software Company Winners](#), [Emerging Tech: 3 Form Factors Will Transform Advanced Wearable Computing](#), [Emerging Tech: Sensors and GenAI Unlock Innovations in Advanced Wearables](#)

AI processing semiconductor (9 chip types) revenue forecast by Electronic Equipment type (\$M)

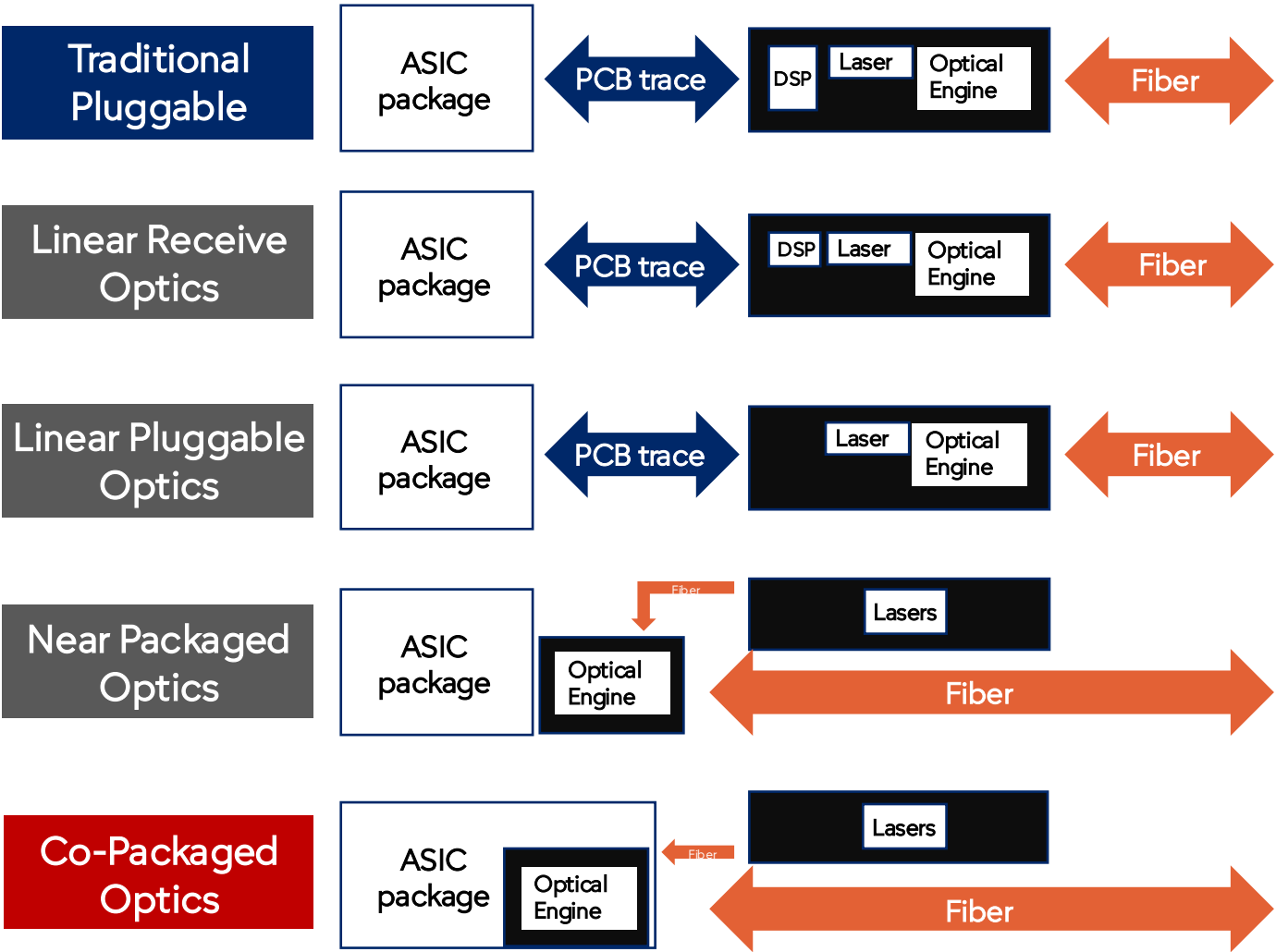
Electronic Equipment Type	2024	2025	2026	2027	2028	2029	2030	2025-30 CAGR
Premium Smartphones	22,212	25,178	25,691	27,814	31,599	32,828	34,602	6.6%
Basic Smartphones	6,663	8,180	7,935	8,476	9,888	11,273	12,152	8.2%
Industrial Smart Robots	30	93	264	508	868	1,450	2,208	88.4%
Smart Watches	692	800	1,005	1,218	1,415	1,672	1,994	20.1%
Video Game Handhelds	0	0	89	257	700	977	947	609.1%
Appliances	32	63	143	291	428	616	903	70.5%
Commercial Drones	76	152	232	314	421	548	678	34.8%
Smart Speakers	126	161	193	262	389	520	666	32.8%
Personal Robots	41	65	104	151	273	458	665	59.0%
Consumer Drones	9	14	37	75	128	180	295	83.7%
Hearables	10	20	37	64	102	156	285	71.0%
Wristbands		1	2	5	13	28	72	157.1%
Grand Total	29,889	34,727	35,731	39,437	46,225	50,706	55,468	9.8%

AI processing Semiconductors categories are:
GPU, AI Accelerator (non-GPU), DSP, Integrated Baseband/Application Processor, MPU (compute and embedded), MCU, FPGA & Programmable Logic Devices

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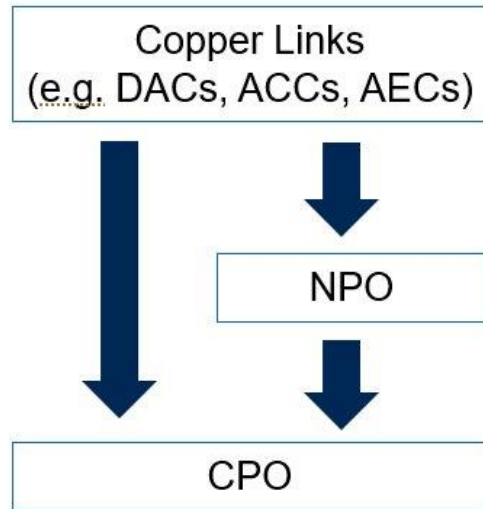
Increasing data rates drive five categories of optics integration

- A common misperception
 - pluggables can't handle the increasing data rates
 - they are power inefficient and limit port density
 - so everything will shift to CPO (co-packaged optics)
- The roadmap is not simple, and it will not flow in a purely logical way

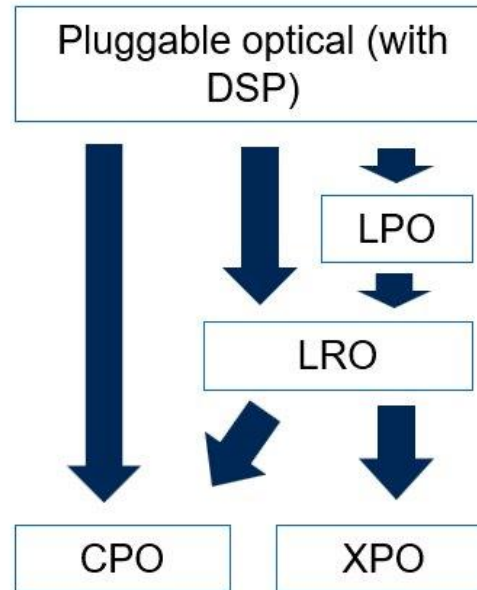


Scale-up and scale-out will take different paths to CPO

Scale-up



Scale-out



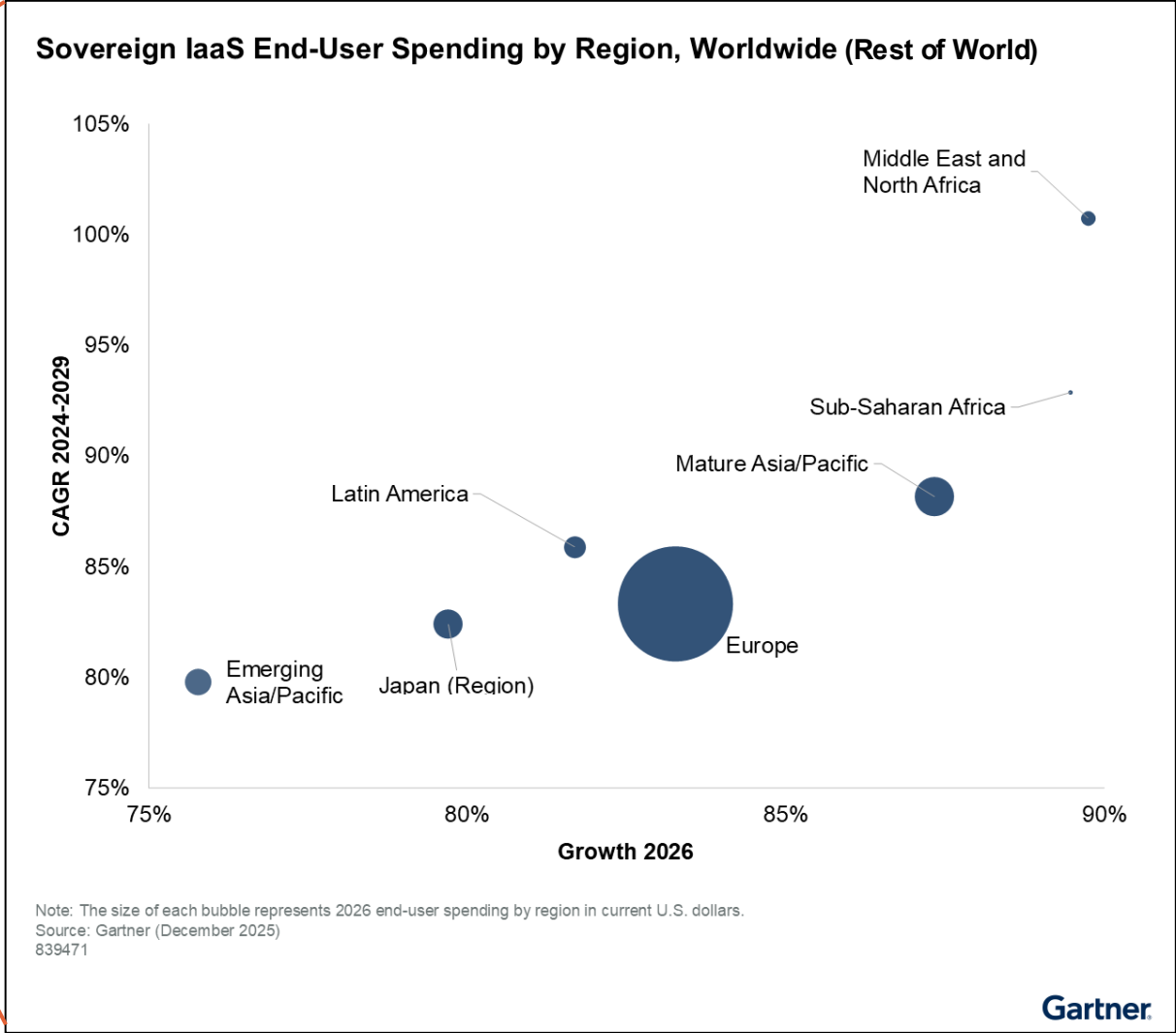
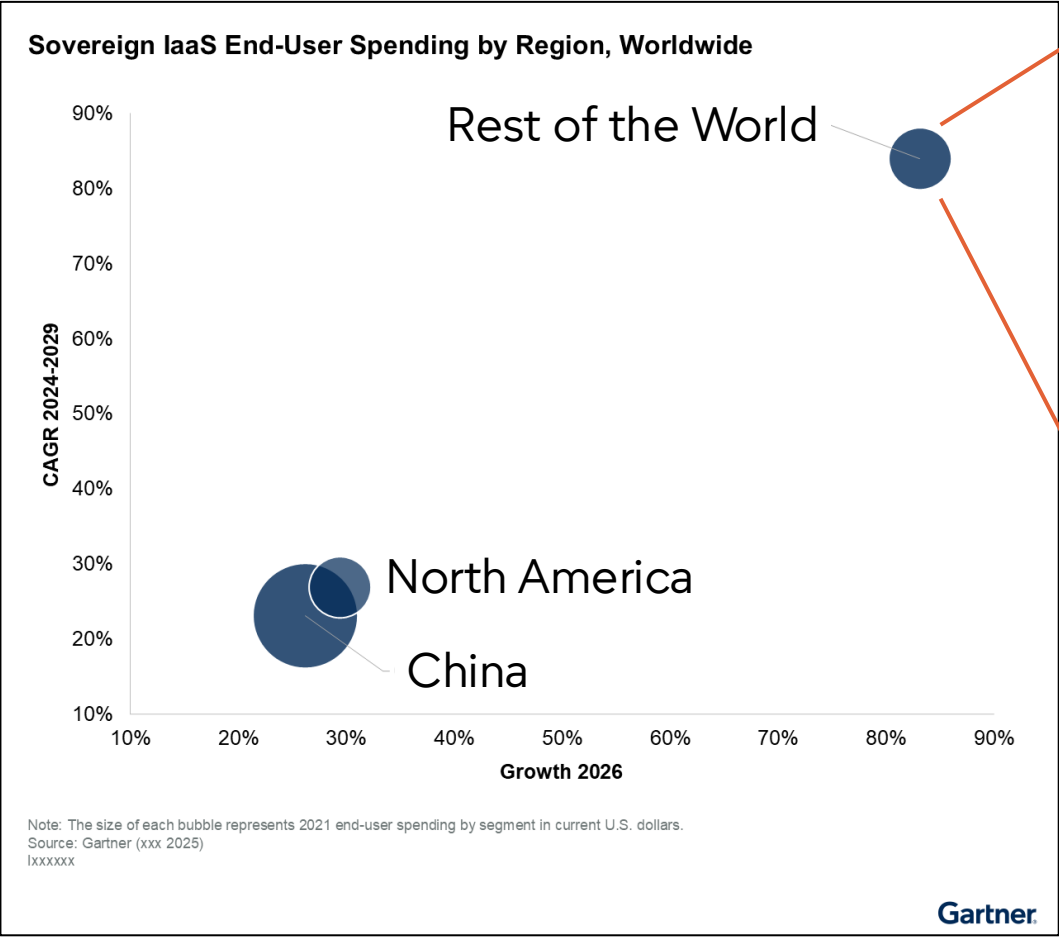
LRO: Linear receive optics
LPO: Linear pluggable optics
NPO: Near-packaged optics
CPO: Co-packaged optics

CPO is still nascent, but by 2029 it will become the go-to connectivity for both scale-up and scale-out interconnects

- Future innovations will provide opportunities across transceiver types

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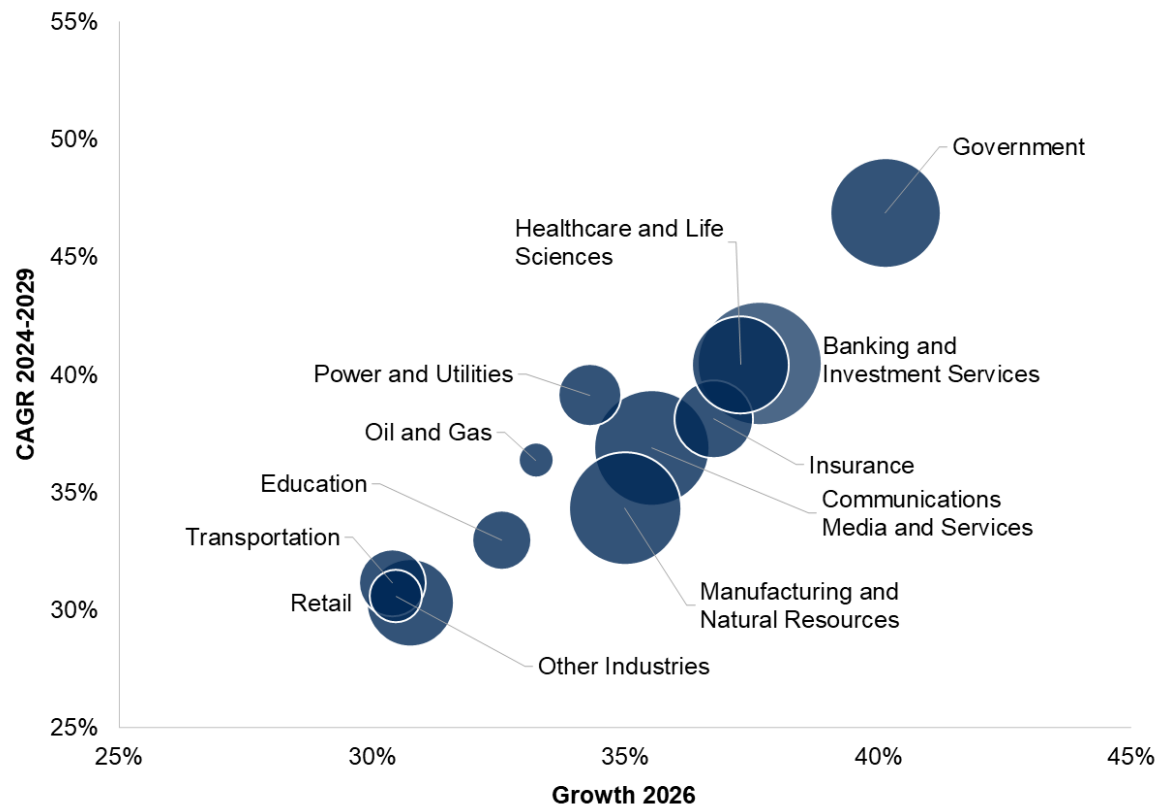
Sovereign AI Priorities go Beyond North America and China



Sovereign Cloud Infrastructure as a Service (IaaS) market will grow to \$224 billion in 2029 (CAGR 2024-29 = 38%)

By 2028, >75% of all enterprises outside of the U.S. will have a digital sovereignty strategy

Sovereign IaaS End-User Spending by Verticals, Worldwide



Note: The size of each bubble represents 2026 end-user spending by verticals in current U.S. dollars.
Source: Gartner (December 2025)
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- Many commercial organizations are increasingly impacted by digital geopolitical factors:
 - Trade restrictions
 - Legislation that impacts global operations
 - Government-imposed restrictions on the acquisition and use of digital technology
- Traditional methods (e.g. service contracts with penalty clauses) don't provide protection from these risks

Recommendations

1. Focus on understanding what AI native electronic equipment will look like for your chosen electronic device, and remember, new form factors will arrive.
2. Prepare for a fragmented optical connectivity market and prepare for different solutions for scale-up and scale-out.
3. Think beyond the U.S. and China Sovereign AI ecosystems and remember that enterprises will provide significant opportunities.



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